JOSÉ DIEGO GONZÁLEZ FERNÁNDEZ

Mechanical Engineering

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PROFILE

Mechanical Engineering student with experience in mechanical design, structural FEA simulation, embedded systems, and telemetry: 3D CAD modeling and assembly in SolidWorks, C/C++ firmware for ESP32 with MQTT, PCB design in Altium, and data processing in Python. Team Captain of a 30-member team focused on developing an energy efficient EV.

EDUCATION

B.Sc. Mechanical Engineering

Tecnológico de Monterrey (ITESM)

August 2024 – June 2028 Mexico City, MX

- Merit Scholarship (60%)
- GPA: 96/100

ENGINEERING EXPERIENCE

Team Captain, Telemetry Leader & Race Engineer

EcoVolt CCM EV Racing Team (Shell Eco-Marathon)

October 2024 – Ongoing Mexico City, MX

- Led a ∼30-member team focused on developing an energy efficient EV, achieving ∼200 km/kWh at a speed of ∼28 km/h.
- Contributed to the manufacturing and assembly of a 100% carbon fiber monocoque chassis, optimizing structural rigidity and overall weight reduction.
- Engineered and deployed a real-time telemetry system to collect and analyze vehicle data, including: GPS, Accelerometer, Gyroscope, Current & Voltage Sensor, Hall-Sensors for speed and throttle/brake inputs.
- Co-designed 4 PCBs in Altium Designer: 3 Telemetry PCBs (including an energy reserve), 1 Hall-Sensors PCB for the motor.

Visiting Student - Micro & Nano fabrication

Massachusetts Institute of Technology (MIT.nano Lab)

July 2026 (40 hours) Cambridge, USA

- **Photovoltaic Cell Microfabrication:** Explored the cleanroom fabrication process for p-n junction silicon solar cells, performing film deposition, dry etching, metallization, and I-V characterization.
- **MEMS & Microfluidic Devices:** Fabricated silicon nitride cantilevers and PDMS microfluidic channels using photolithography, chemical etching, and soft lithography.

PUBLICATIONS

Real-Time Telemetry and Computational Intelligence Pipeline for Energy-Efficient Driving of an Electric Eco-Marathon Vehicle

IEEE SAIBOT 2026 (Latin American School on Computational Intelligence and Robotics) – *submitted*

September 2026

P. L. Ruíz-Servín, **J.D. González-Fernández**, A. Flores-Mendoza, J. Izquierdo-Reyes

VOLUNTEERING

Mentoring

Tecnológico de Monterrey (ITESM)

August 2023 – Ongoing

- Provided comprehensive tutoring for IB Mathematics and first-year university students, helping students clarify complex concepts, develop effective study strategies, and improve exam readiness.

CERTIFICATIONS

- Solidworks CSWA-Simulation (2026)
- Solidworks CSWP (2026)
- Altium: PCB Basic Design (2026)
- MIT.nano: Cert. of Participation (2026)
- Solidworks CSWA (2025)

AWARDS

- Data & Telemetry Award Runner-Up Shell Eco-Marathon Americas (2026)
- Honorable mention Altair Global Student Contest (2025)
- Data & Telemetry Award Winner Shell Eco-Marathon Americas (2025)

SKILLS

Programming:

Python | C/C++ | LABVIEW

Frameworks/IDEs:

Git | Visual Studio Code | Platformio

Design & Simulation:

Solidworks CAD/Simulation | OnShape | Fusion CAM | MATLAB/Simulink

Manufacturing & Prototyping

3D Printing | Lathe | Milling | CNC | Welding | SMD/THT soldering

LANGUAGES

Spanish	Native
English	B2/C1
French	B1/B2
German	A2
Italian	A2